

Automated, on-demand Sentinel imagery retrieval through a service-based workflow

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Under the framework of the AMMASO project, the tool presented in this research functions as a delivery system of satellite imagery to be used in air and water pollution applications, and as the main component of a workflow for object identification in aquatic environments. This work is based on the Copernicus Dataspace Ecosystem capabilities, hosting a variety of Earth Observation services, tools and infrastructure. For the purposes of this research, the Sentinel Hub service was used which doubles as a one-stop shop for satellite imagery in terms of spectral, spatial and temporal resolution, and territorial accessibility. Sentinel-1 and Sentinel-2 imagery was considered essential and thus the workflow is constructed around these products.

This tool is based on the Sentinel Hub's examples, and it constitutes a more detailed evalscript query for operational usage, but unlike the steps followed for a basic evalscript this approach eliminates hardcoding wherever possible. From an accessibility aspect the main goal, is to pass control to the user that might not be familiar with the Sentinel Hub infrastructure or a user that demands easy, time efficient, and seamless access. This is achieved by the ability of the tool to provide essential query arguments customisation by featuring configurable parameters. Performance-wise, as there are processing unit limits, the script is set up as so to avoid resource wastage by optimally employing parameter filtering. Overall, the service developed allows for automated on-demand retrieval of satellite imagery with emphasis on the aquatic environment and is expected to support research initiatives related to the shipping identification and emissions estimation field.